An IoT-Based Intelligent Drainage Monitoring and Early Blockage Detection System

1. Methodology

Urban drains are often partially blocked by plastic, leaves, mud and stones, especially during heavy rainfall. Conventional drains give little early warning, and overflow becomes visible only after the flow is already restricted. The proposed system provides continuous, low-cost monitoring and an early warning of possible blockage.

A rain sensor first detects rainfall. An ESP32 microcontroller then starts drainage monitoring and collects readings from two water-level sensors placed inside the drain: Sensor A on the upstream side and Sensor B on the downstream side. During normal flow, the two readings remain relatively close. During a possible blockage, water accumulates upstream while the downstream level stays comparatively lower. The ESP32 calculates the water-level difference **D = A − B** and evaluates it together with the rainfall condition. A three-level decision system classifies the drain as **Normal, Warning or Blockage Alert**. LEDs and a buzzer give local indication, while Wi-Fi sends a remote notification.

Innovation: The system does not depend on a single high-water-level threshold. Two-point differential monitoring combined with rainfall context detects abnormal drainage behaviour earlier and reduces false alarms caused by ordinary water-level fluctuations. The modular design can be extended with additional sensor nodes for other drain sections.

2. Proposed Block Diagram

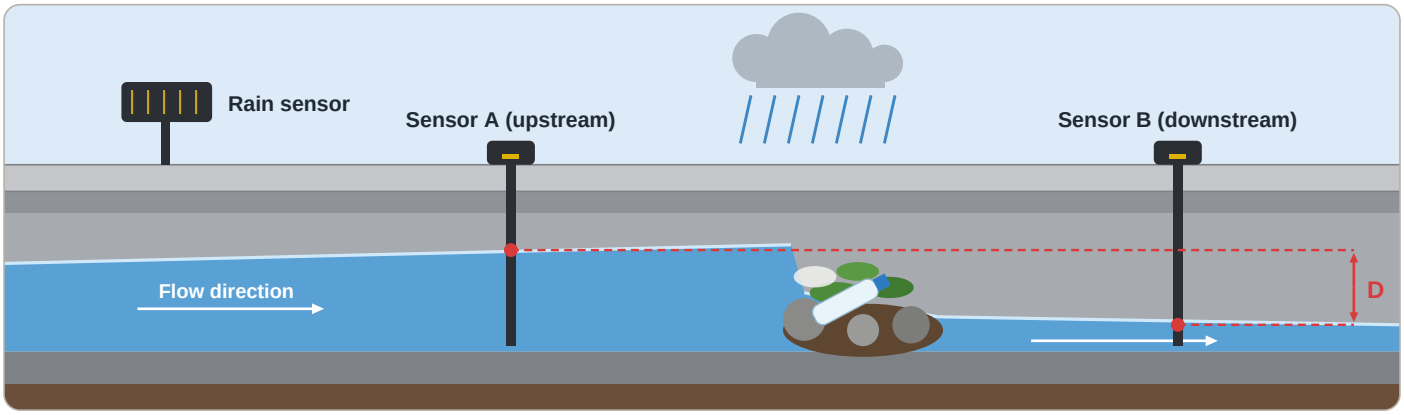


Fig. 1 Two-point water-level monitoring inside a drainage channel (D = A − B)

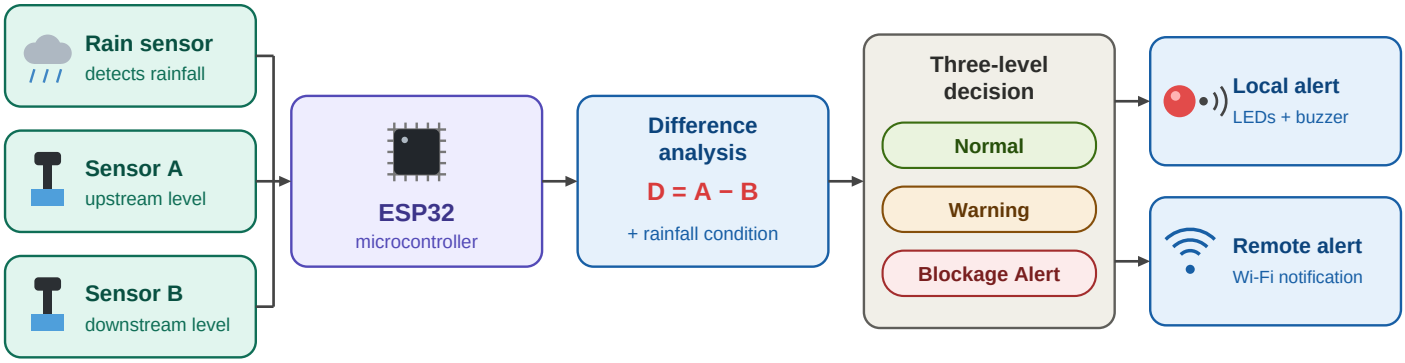


Fig. 2 Block diagram: continuous monitoring → abnormal-flow decision → early warning

3. Proposed Outcome

A working small-scale prototype that displays real-time drainage status, identifies abnormal upstream/downstream water-level differences during rainfall, and generates an early warning of possible blockage through local alerts and Wi-Fi (IoT) notification. The system is intended to support preventive inspection and faster response before severe overflow occurs.